

Metallographic preparation of stainless steel

Corrosion resistant steels contain at least 11% chromium and are collectively known as “stainless steels”. Within this group of high alloy steels four categories can be identified: ferritic, martensitic, austenitic, and austenitic-ferritic (duplex) stainless steels. These categories describe the alloy's microstructure at room temperature, which is largely influenced by the alloy composition.

The main characteristic of stainless steels is their corrosion resistance. This property can be enhanced by the addition of specific alloying elements, which have a further beneficial effect on other characteristics such as toughness and oxidation resistance.

For instance, niobium and titanium increase resistance against intergranular corrosion as they absorb the carbon to form carbides; nitrogen increases strength and sulphur increases machinability, because it forms small manganese sulphides which result in short machining chips. Due to their corrosion resistance and superior surface finishes stainless steels play a major part in the aircraft,



High performance stainless steel parts for the aircraft industry

chemical, medical and food industries, in professional kitchens, architecture and even jewelry.

Metallography of stainless steels is an important part of the overall quality control of the production process. The main metallographic tests are grain size measurement, identification of delta ferrite and sigma phase and the evaluation and distribution of carbides. In addition, metallography is used in failure analysis investigating corrosion/oxidation mechanisms.



Fig. 1: Duplex steel etched electrolytically with 40% aqueous sodium hydroxide solution, showing blue austenite and yellow ferrite 150x

Difficulties during metallographic preparation

Grinding and polishing: Deformation and scratching in ferritic and austenitic stainless steels. Retention of carbides and inclusions.



Surface of stainless steel after 3µm polish, showing deformation from grinding DIC 25x



Insufficiently polished stainless steel after colour etching (Beraha II), showing deformation 100x

Solution:

Thorough diamond polishing and final polishing with colloidal silica or alumina.



Production and application of stainless steel

The production process of high alloy steels is a sophisticated process of melting and remelting. A mixture of iron and well sorted scrap is first melted in an electric arc furnace and then cast into ingot form or continuously cast into bloom or billet. For many applications these primary products can be further processed into bar, rod or plate form. For steels with higher quality demands, the primary product can be used as feedstock for a secondary steelmaking process. This secondary process can be a double or even triple remelt by vacuum induction melting plus vacuum arc remelting or electroslag remelting, which can also be done under pressure and protective gases.

The main purpose of this secondary process is to reduce impurities such as oxides, sulphides and silicates so that with successive remelts the degree of cleanliness increases and homogenous ingots with excellent mechanical and physical properties are produced.

Application

The corrosion resistance of stainless steels is based on alloying chromium with iron, and is dependant on the formation of a passive surface oxide layer, which rebuilds itself spontaneously when mechanically damaged.

A variety of different types of corrosion can occur, such as pitting, stress, intercrystalline or vibrational corrosion. Improved resistance against any specific form of attack can be provided by adding alloying elements other than chromium, for instance molybdenum, which improves resistance against pitting corrosion. The main alloys, properties and examples of applications of the four types of stainless steels are briefly described:

Ferritic stainless steels: are non heat treatable alloys with a low carbon content and 11-17% chromium.

Properties: magnetic, resistant to atmospheric corrosion, moderate strength and toughness.

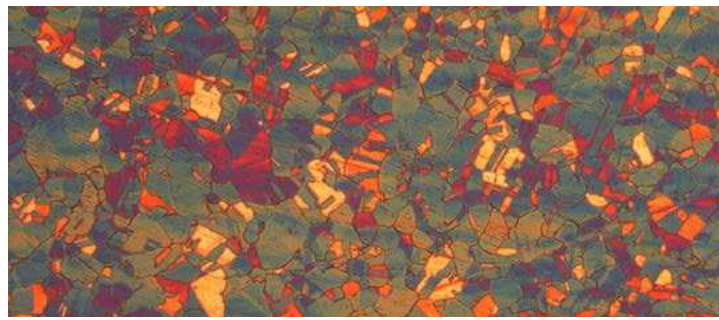
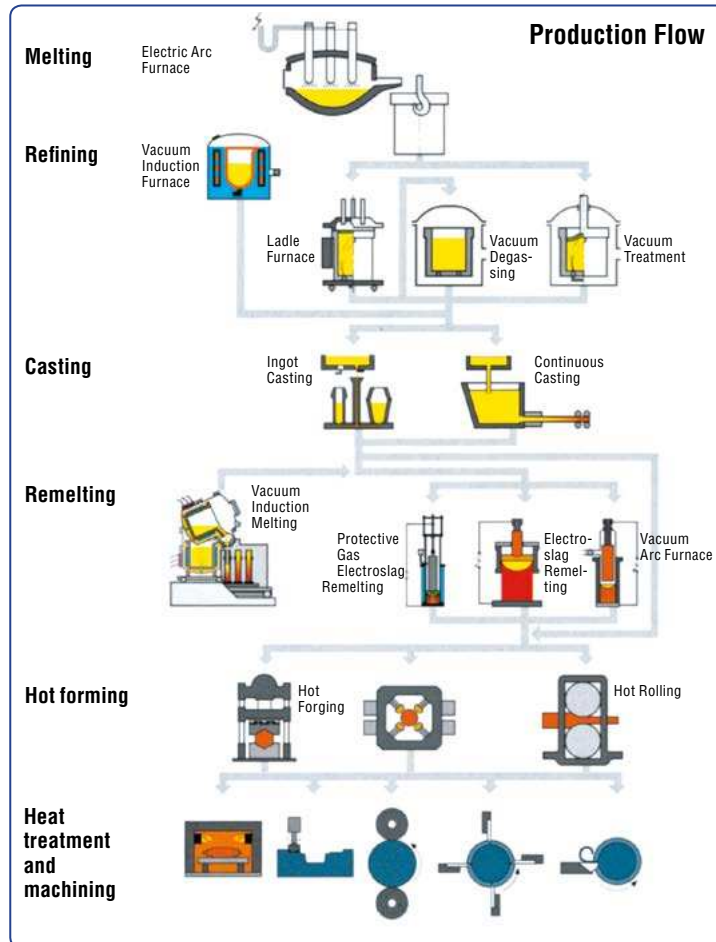


Fig.2: Austenitic steel, colour etched (Beraha II).

100x



Applications: magnetic valves, razor blades, car trim.

Martensitic stainless steels are heat treatable alloys with medium carbon content, 12-18% chromium and 2-4% nickel. Properties: high corrosion resistance, and high temperature and creep resistance. Applications: scalpels, knives, hooks and tweezers in medical applications, drive systems and high performance parts for airplanes.

Austenitic stainless steels are not heat treatable, have 0.03-0.05 % carbon, main alloying elements are chromium (17-24 %), nickel (8-25%) and molybdenum (2-4%); titanium and niobium are added for carbide forming. Properties: high ductility, high corrosion resistance, resistant to oxidizing acids, alkalis, very good cold forming properties, easy to work and machine.

Applications: screws, bolts and implants, low temperature applications, vessels and pipes in the chemical, pharmaceutical and food industries, kitchen utensils.

Austenitic-ferritic steels, (Duplex) have a low carbon content and generally higher chromium (21-24%) and lower nickel content (4-6%) than austenitic steels, and 2-3% molybdenum.

Properties: fatigue resistance in corrosive media, good resistance against stress corrosion.

Applications: equipment for chemical, environmental and offshore industries, architecture.



Difficulties in the preparation of stainless steels

Ferritic stainless steels are soft and austenitic steels are ductile. Both are prone to mechanical deformation. Final polishing usually leaves these steels highly reflective, however, if they are not thoroughly prepolished, deformation can reappear after etching (Fig. 3).

Due to their hardness, martensitic steels are relatively easy to polish. In general, care should be taken to preserve the carbides.

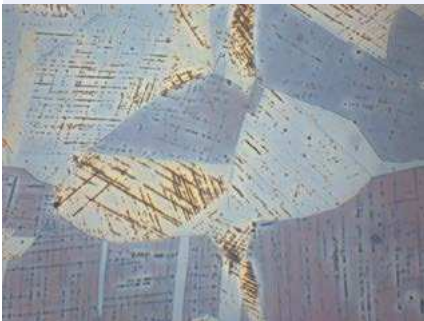


Fig. 3: Austenitic steel insufficiently polished showing deformation after etching (Beraha II) 500x

Recommendations for the preparation of stainless steels

It is strongly recommended that for the soft and ductile stainless steels the use of very coarse grinding foil/papers and high pressures should be avoided, as this can result in deep deformation. As a general rule, the finest possible grit, consistent with the sample area and surface roughness, should be used for plane grinding. Fine grinding is carried out with diamond on a rigid disc (MD-Largo) or, as an alternative for some types of stainless steels, on a MD-Plan cloth. Fine grinding is followed by a thorough diamond polish on a medium soft cloth, and the final polish with colloidal silica (OP-S), or alumina (OP-A) removes the fine scratches. This step should be very thorough and it can take up to several minutes. A good final polish increases the chance for a better contrast (see "Etching").

Any deformation from the first grinding step, which is not removed by fine grinding, will leave its traces and can not be removed by final polishing.

Table 1 shows a preparation method for stainless steel samples, 30 mm mounted, on the semi-automatic Tegramin 300 mm diameter. Table 2 shows a preparation method for 6 stainless steel samples, 65x30 mm, cold mounted or unmounted using Struers MAPS or AbraPlan/AbraPol, 350 mm diameter.

Electrolytic polishing

For research work or fast, general structure check, electrolytical polishing and etching can be an alternative to mechanical polishing of stainless steels, as it does not leave any mechanical deformation.

Electrolytic polishing gives excellent results for checking the microstructure (Fig. 4), but it is not suited to identify carbides. They are washed out or appear enlarged.

Before electrolytic polishing, the samples have to be ground to 1000# on silicon carbide foil/paper. The finer the initial surface the better the results of the electrolytical polish (see preparation method below).

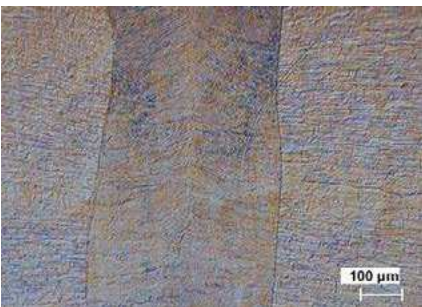


Fig. 4: Stainless steel weld, polished and etched electrolytically, DIC

Electrolyte:	A3
Area:	1cm ²
Voltage:	35 V
Flowrate:	13
Time:	25 sec
External etching with stainless steel etching dish:	
10% aqueous oxalic acid	
Voltage:	15V
Time:	60 sec

Preparation method for electrolytic polishing and etching of stainless steel. Grinding on SiC foil/paper 320#, 500# and 1000#.

Grinding

Step		PG	FG
	Surface	SiC-Paper 220	MD-Largo
	Abrasive	Type	SiC
		Size	#220
	Suspension/Lubricant	Water	DiaPro Allegro/Largo 9
	rpm	300	150
	Force [N]/specimen	25	40
	Time (min)	As needed	5

Polishing

Step		DP	OP
	Surface	MD-Dac	MD-Chem
	Abrasive	Type	Diamond
		Size	3 µm
	Suspension/Lubricant	DiaPro Dac 3	OP-S NonDry OP-A
	rpm	150	150
	Force [N]/specimen	20	15
	Time (min)	4	2-3

Table 1: Preparation method for stainless steel samples, 30 mm diameter mounted, on the semi-automatic Tegramin, 300 mm diameter.

As an alternative to DiaPro polycrystalline diamond suspension P, 9 µm, 3 µm and 1 µm can be used together with green/blue lubricant.

Etching

Etching stainless steels requires some experience and patience. The literature for etchants is extensive, and it is recommended to try a variety in order to set up an individual stock of solutions appropriate for the particular material which is regularly prepared in the laboratory.

By virtue of the fact that stainless steels are highly corrosion resistant, very strong acids are required to reveal their structure. Standard safety precautions have to be used when handling these etchants. In many laboratories the etchants mentioned in the literature will be modified according to the material to be etched or even out of personal preference. For good etching results, a sufficient final oxide polishing is essential. Following are some etchants which have proved successful in every day, routine applications.

Grinding

Step		PG	FG
	Surface	Stone 3A36	MD-Largo
	Abrasive	Type	Al ₂ O ₃
		Size	#150
	Suspension/ Lubricant	Water	DiaPro Allegro/Largo 9
	rpm	1450	150
	Force [N]/ specimen	50	50
	Time (min)	As needed	9



Polishing

Step		DP 1	DP 2	OP
	Surface	MD-Mol APS	MD-Nap	MD-Chem
	Abrasive	Type	Diamond	Silica/Alumina
		Size	3 µm	0.04/0.02 µm
	Suspension/ Lubricant	DiaPro Mol B 3	DiaPro Nap B 1	OP-S NonDry OP-A
	rpm	150	150	150
	Force [N]/ specimen	50	25	25
	Time (min)	6	4	2-3

Table 2: Preparation method for stainless steel samples, 65x30mm, cold mounted or unmounted using Struers MAPS or AbraPlan/AbraPol, 350 mm diameter.

As an alternative to DiaPro polycrystalline diamond suspension P, 9 µm, 3 µm and 1 µm can be used together with green/blue lubricant.

Chemical etching

Caution: Always follow the recommended safety precautions when working with chemical reagents.

For martensitic steels

- 925 ml ethanol
- 25 g picric acid
- 50 ml hydrochloric acid

For austenitic steels

- 1) Swab etch:
 - 500 ml distilled water
 - 300 ml hydrochloric acid
 - 200 ml nitric acid
 - 50 ml of a saturated iron-III-chloride solution
 - 2.5 g copper-II-chloride
- 2) 100 ml water
- 300 ml hydrochloric acid
- 15 ml hydrogen peroxide (30%)

- 3) V2A etchant:
 - 100 ml water
 - 100 ml hydrochloric acid
 - 10 ml nitric acid
 Etch at room temperature or up to 50°C

Color etchant Beraha II:

- Stock solution
 - 800 ml distilled water
 - 400 ml hydrochloric acid
 - 48 g ammonium bifluoride
- To 100 ml of this stock solution add 1-2 g potassium metabisulfite for etching.

Electrolytic etching

- For austenitic-ferritic steels (Duplex)
 - 40% aqueous sodium hydroxide solution

All stainless steels:

- 10% aqueous oxalic acid

Structure interpretation

Ferritic stainless steels do not respond to heat treatment. Their properties however can be influenced by cold working. They are magnetic at room temperature. The microstructure in the annealed condition consists of ferrite grains in which fine carbides are embedded. Ferritic steels used for machining contain a large amount of manganese sulfides to facilitate free cutting (Fig. 5).

Martensitic stainless steels respond to heat treatment. Martensite is formed through rapid cooling and properties can then be optimised by subsequent tempering treatment. The alloys are magnetic. Depending on the thermal treatment the microstructure can range from pure martensitic structure to fine tempered martensite. Different alloys and various dimensions of semi-finished products require complex heat treatment temperatures and times.

Delta ferrite (Fig. 6) is usually an unwanted phase, because long annealing times of steels with high chromium content, at temperatures between 700 and 950°C, can change the delta ferrite into the hard and brittle iron-chromium intermetallic sigma phase.

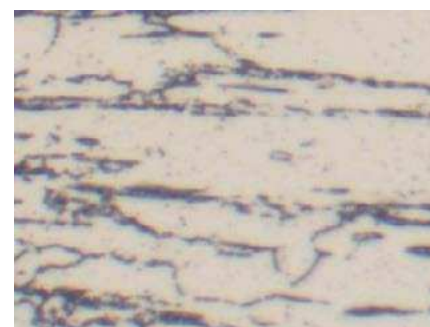


Fig. 5: Ferritic stainless steel with manganese sulfides and strings of small carbides, etched electrolytically with 10% oxalic acid 200x



Fig. 6: Tempered martensitic stainless steel with delta ferrite, etched with picric acid 75x

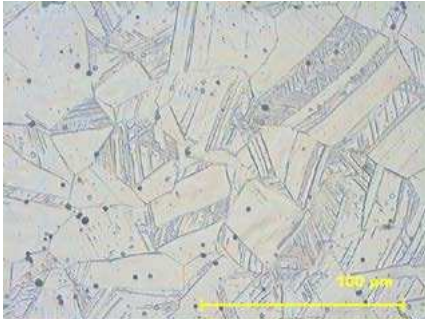


Fig. 7: Cold worked austenitic steel showing twinning, etched with V2A etchant.



Fig. 8: Austenite with carbides and some titanium carbon nitrides 200x



Fig. 9: Austenitic steel with strings of delta ferrite, showing microsegregations. Blue areas: depletion of alloying elements. 125x

Heating up to 1050°C and subsequent quenching removes the sigma phase and with it the embrittlement.

Austenitic stainless steels do not respond to thermal treatment, instead, rapid cooling results in the production of their softest condition. In this state they are non-magnetic and their properties are influenced by cold working. The microstructure of these steels consists of austenite grains which may exhibit twinning (Fig. 7). Exposure of these steels to elevated temperatures in the region of 600-700°C can result in the formation of

complex carbides within the austenite grains. This leads to an impoverishment of chromium in the austenite solid solution, which increases the susceptibility to intergranular corrosion or oxidation.

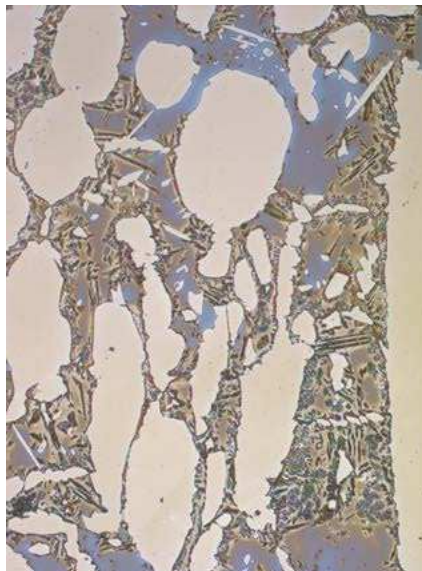
By reducing the carbon to below 0.015% and adding small amounts of titanium or niobium, the risk of intergranular corrosion is reduced, as these elements form carbides in preference to the chrome (Fig. 8).

Delta ferrite can appear due to critical heat treatment conditions in martensitic or cold working of austenitic steels (Fig. 9).

Austenitic-ferritic stainless steels (Duplex) consist of ferrite and austenite. Electrolytic etching in a 40% caustic soda solution reveals the structure and the correct percentage of each phase can be estimated (see Fig.1, and Fig. 10 below). These steels are ductile and are specifically used in the food, paper and petroleum industries.

Summary

Stainless steels are corrosion resistant steels with high chromium and nickel contents. Ferritic and stainless steels are soft, respectively ductile, and are prone to mechanical deformation and scratching during metallographic preparation. In addition, carbides can not always be retained.



For a successful mechanical polish, it is suggested that

- coarse abrasives for plane grinding are avoided.
- fine grinding and polishing with diamond should be thorough and ensure removal of all deformation from plane grinding.
- a final oxide polish with colloidal silica or alumina should be carried out to provide a deformation free surface.

A four step procedure, completed on automatic preparation equipment, gives good and reproducible results. Stainless steels are difficult to etch chemically and the recommended etchants are very corrosive and require careful handling.

Alternatively, electrolytical polishing and etching is recommended, which gives a deformation free surface, but does not retain carbides.

Fig.10: Forged duplex steel showing blue ferrite, white austenite and fine needles of sigma phase, etched electrolytically with 40% aqueous sodium hydroxide.

150x



Struers A/S
Pederstrupvej 84
DK-2750 Ballerup, Denmark
Phone +45 44 600 800
Fax +45 44 600 801
struers@struers.dk
www.struers.com

Application Notes

Metallographic preparation of stainless steel

Elisabeth Weidmann, Struers A/S, Copenhagen
Anne Guesnier, Struers A/S, Copenhagen
Bill Taylor, Struers, Ltd., Glasgow, UK.

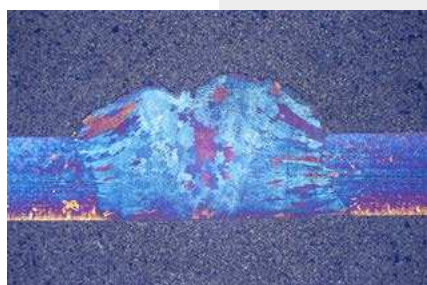
Acknowledgements

We wish to thank Böhler Edelstahl GmbH, Kapfenberg, Austria, for generously supplying information, sample material, and the permission to reproduce the photo of parts on page 1 and the diagram of production flow on page 2. Special thanks go to J. Hofstätter for the co-operation and to A. Dreindl for contributing micrographs Figs. 1, 2, 6, 9 and 10.

We thank Dr. H. Schnarr, Struers GmbH, Willich, Germany, for micrographs Figs. 4 and 7.

Bibliography

Schumann, VEB Deutscher Verlag für Grundstoffindustrie, Leipzig 1968
Werkstoffkunde und Werkstoffprüfung, W. Domke, Verlag W. Giradet, Essen, 1977
Metals Handbook, Desk Edition, ASM, Metals Park, Ohio, 44073, 1985
Color Metallography, E. Beraha, B. Shpigler, ASM, Metals Park, Ohio, 44073, 1977
Handbuch der metallographischen Ätzverfahren, M. Beckert, H. Klemm, VEB Deutscher Verlag für Grundstoffindustrie, Leipzig, 1976
Metallography, Principles and Practice, George F. VanderVoort, McGraw-Hill Book Company, 1984
Merkblatt 821, Edelstahl Rostfrei-Eigenschaften Informationsstelle Edelstahl Rostfrei, Düsseldorf, BRD
Metallographic instructions for color etching by immersion, E. Weck, E. Leistner, Deutscher Verlag für Schweisstechnik (DVS), Düsseldorf, 1983



Stainless steel weld etched according to Beraha II.

20x

AUSTRALIA & NEW ZEALAND

Struers Australia
27 Mayneview Street
Milton QLD 4064
Australia
Phone +61 7 3512 9600
Fax +61 7 3369 8200
info.au@struers.dk

BELGIQUE (Wallonie)
Struers S.A.S.
370, rue du Marché Rollay
F-94507 Champigny
sur Marne Cedex
Téléphone +33 1 5509 1430
Télécopie +33 1 5509 1449
struers@struers.fr

BELGIUM (Flanders)
Struers GmbH Nederland
Elektraweg 5
3144 CB Maassluis
Telefoon +31 (10) 599 7209
Fax +31 (10) 599 7201
netherlands@struers.de

CANADA
Struers Ltd.
7275 West Credit Avenue
Mississauga, Ontario L5N 5M9
Phone +1 905-814-8855
Fax +1 905-814-1440
info@struers.com

CHINA
Struers Ltd.
No. 1696 Zhang Heng Road
Zhang Jiang Hi-Tech Park
Shanghai 201203, P.R. China
Phone +86 (21) 6035 3900
Fax +86 (21) 6035 3999
struers@struers.cn

CZECH REPUBLIC
Struers GmbH
Organizační složka
Havlíčkova 361
CZ-252 63 Roztoky u Prahy
Phone +420 233 312 625
Fax +420 233 312 640
czechrepublic@struers.de

DEUTSCHLAND
Struers GmbH
Carl-Friedrich-Benz-Straße 5
D-47877 Willich
Telefon +49 (0) 2154 486-0
Fax +49 (0) 2154 486-222
verkauf@struers.de

FRANCE
Struers S.A.S.
370, rue du Marché Rollay
F-94507 Champigny
sur Marne Cedex
Téléphone +33 1 5509 1430
Télécopie +33 1 5509 1449
struers@struers.fr

HUNGARY
Struers GmbH
Magyarországi Fióktelep
Tatai út 53
2821 Gyermely
Phone +36 (34) 880546
Fax +36 (34) 880547
hungary@struers.de

IRELAND
Struers Ltd.
Unit 11 Evolution @ AMP
Whittle Way, Catcliffe
Rotherham S60 5BL
Tel. +44 0845 604 6664
Fax +44 0845 604 6651
info@struers.co.uk

ITALY
Struers Italia
Via Monte Grappa 80/4
20020 Arese (MI)
Tel. +39-02/38236281
Fax +39-02/38236274
struers.it@struers.it

JAPAN
Marumoto Struers K.K.
Takara 3rd Building
18-6, Higashi Ueno 1-chome
Taiko-ku, Tokyo 110-0015
Phone +81 3 5688 2914
Fax +81 3 5688 2927
struers@struers.co.jp

NETHERLANDS
Struers GmbH Nederland
Elektraweg 5
3144 CB Maassluis
Telefoon +31 (10) 599 7209
Fax +31 (10) 599 7201
netherlands@struers.de

NORWAY
Struers A/S, Norge
Sjøslogenveien 44C
1407 Vinterbro
Telefon +47 970 94 285
info@struers.no

ÖSTERREICH
Struers GmbH
Zweigniederlassung Österreich
Betriebsgebiet Puch Nord 8
5412 Puch
Telefon +43 6245 70567
Fax +43 6245 70567-78
austria@struers.de

POLAND
Struers Sp. z o.o.
Oddział w Polsce
ul. Jasnogórska 44
31-358 Kraków
Phone +48 12 661 20 60
Fax +48 12 626 01 46
poland@struers.de

ROMANIA
Struers GmbH
Sucursala Sibiu
Str. Scoala de Inot, nr. 18
RO-550005 Sibiu
Phone +40 269 244 558
Fax +40 269 244 559
romania@struers.de

SCHWEIZ
Struers GmbH
Zweigniederlassung Schweiz
Weissenbrunnstraße 41
CH-8903 Birmensdorf
Telefon +41 44 777 63 07
Fax +41 44 777 63 09
switzerland@struers.de

SINGAPORE
Struers Singapore
627A Aljunied Road,
#07-08 BizTech Centre
Singapore 389842
Phone +65 6299 2268
Fax +65 6299 2661
struers.sg@struers.dk

SPAIN
Struers A/S, Spain
Camino Cerro de los Gamos 1
Building 1 - Pozuelo de Alarcón
CP 28224 Madrid
Téléfono +34 917 901 204
Fax +34 917 901 112
struers.es@struers.es

SUOMI
Struers A/S, Suomi
Hietalahdenranta 13
00180 Helsinki
Puhelin +358 (0)207 919 430
Faksi +358 (0)207 919 431
finland@struers.fi

SWEDEN
Struers Sverige
Ekbacksvägen 22
168 69 Bromma
Telefon +46 (0)8 447 53 90
Telefax +46 (0)8 447 53 99
info@struers.se

UNITED KINGDOM
Struers Ltd.
Unit 11 Evolution @ AMP
Whittle Way, Catcliffe
Rotherham S60 5BL
Tel. +44 0845 604 6664
Fax +44 0845 604 6651
info@struers.co.uk

USA
Struers Inc.
24766 Detroit Road
Westlake, OH 44145-1598
Phone +1 440 871 0071
Fax +1 440 871 8188
info@struers.com